

Cognitive Models, Crowd Wisdom, Citizen Science, and Frogs: An Extension of Brown et al. (2024)

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Detecting Frogs

- People take the role of “citizen scientists” by listening to audio clips trying to detect the presence of seven types of frogs by their calls

environment



person



data



Detecting Frogs

- The same person can listen to multiple clips

environment



person



data



Detecting Frogs

- Multiple people can independently listen to the same clip

environment



person



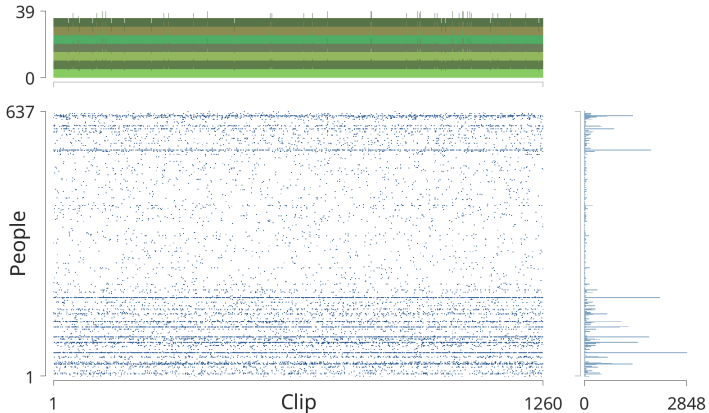
data



Behavioral Data

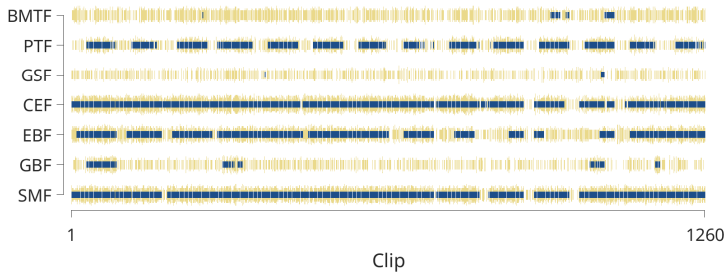
Data

- The joint and marginal distributions of people judging clips shows
 - lots of people judge few clips, but a few people judge many
 - the data collection system aims to have five judgments for each type of frog in each clip



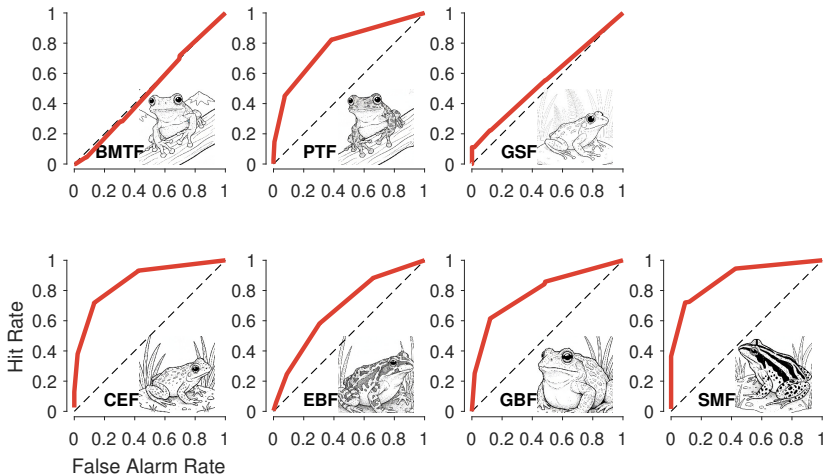
Crowd Proportions and the Truth

- The blue marks indicate the ground truth (expert derived) presence of each type of frog across the sequence of clips
- The yellow bars show the proportion of people who judged a frog was present in each clip



Crowd Performance

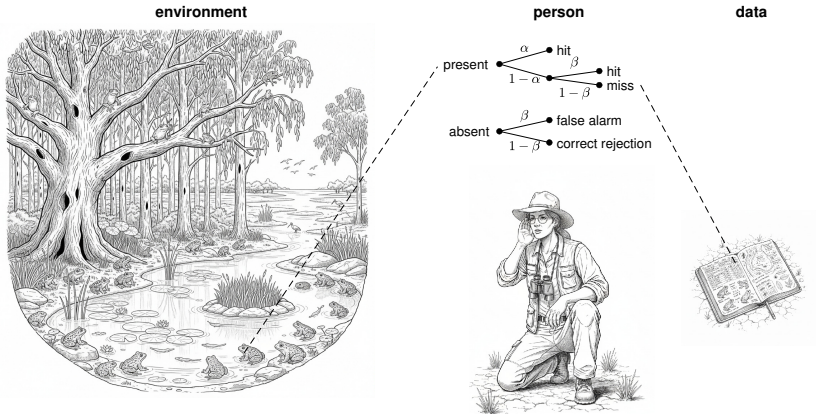
- ROC curves show reasonable wisdom the crowd performance for five of the frog types
 - the BMTF and GSF tree frogs are not well detected
 - both are rarely present but people often report detecting them



Model-Based Wisdom of the Crowd

Model-Based Wisdom of the Crowd

- The model-based approach uses cognitive models to explain how the behavior data were produced (Brown et al., 2024; Lee, 2024)
 - make inferences about the cognitive abilities and biases of individuals
 - make inferences about latent states of nature



One-High Threshold Model

- The model assumes frog f in clip j is either $\tau_{fj} = 1$ present or $\tau_{fj} = 0$ absent with a base-rate that depends on the frog ϕ_f

$$\tau_{fj} \sim \text{Bernoulli}(\phi_j)$$

$$\phi_j \sim \text{uniform}(0, 1)$$

- Individual i judges frog f in clip j to be $y_{ijf} = 1$ present with probability θ_{ijf} given by the one-high threshold model

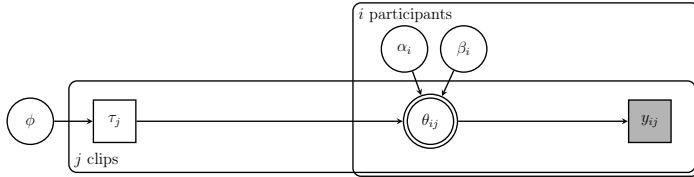
$$\alpha_i, \beta_i \sim \text{uniform}(0, 1)$$

$$\theta_{ijf} = \begin{cases} \alpha_i + (1 - \alpha_i) \beta_i & \text{if } \tau_{fj} = 1 \\ \beta_i & \text{if } \tau_{fj} = 0 \end{cases}$$

$$y_{ijf} \sim \text{Bernoulli}(\theta_{ijf})$$

- Implemented the model in JAGS and Stan

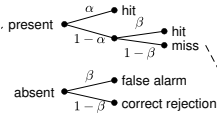
Inferences about both the Stimulus and Cognitive Processes



environment



person

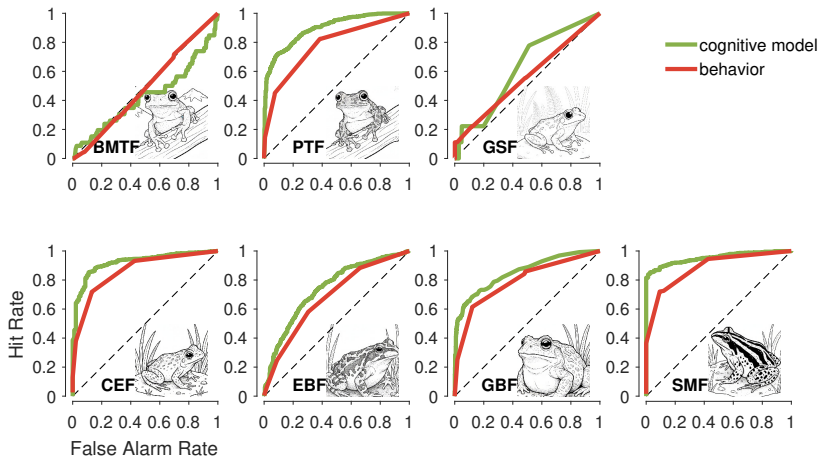


data



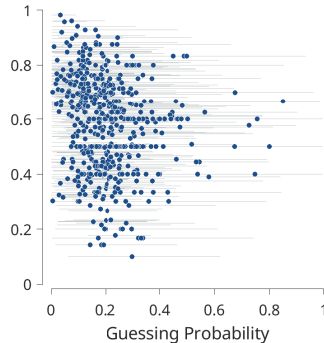
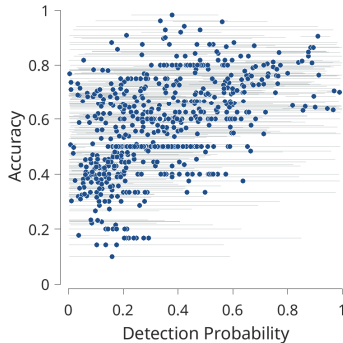
Model-Based Crowd Performance

- The model-based approach performs better than the standard wisdom of the crowd aggregation of behavioral judgments



Inferences about Detection and Guessing Bias

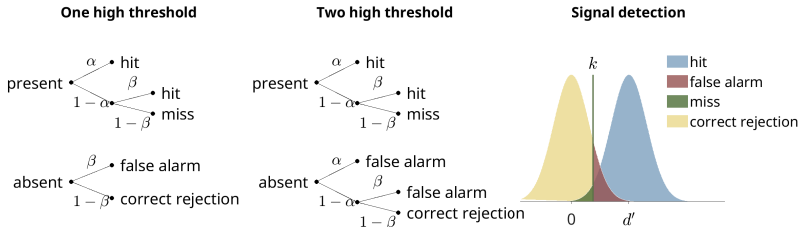
- Inferences about detection ability α_i and guessing bias β_i provide **predictions** about the performance of individual i
 - the accuracy of these predictions allows, for example, “yes” judgments from non-experts to be explained away as guessing



Which Cognitive Model?

Three Cognitive Models

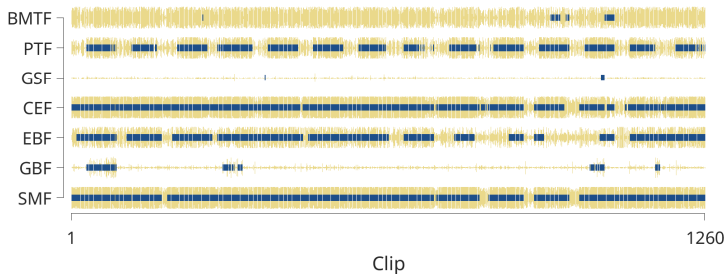
- There are many other possible cognitive models of the detection decisions
 - e.g., the two high threshold model, signal detection models, ...
- They all seem to perform very similarly
 - same ROC curves, and an “ability” parameter that relates to observed accuracy



Modeling the Environment

Model-Based Crowd and the Truth

- Brown et al. (2024) point out that their model-based approach assumes independence over both scientists and clips
 - reasonable assumption for scientists, but there are obvious environmental dynamics that create dependencies over clips
 - “frogs come and frogs go” . . .



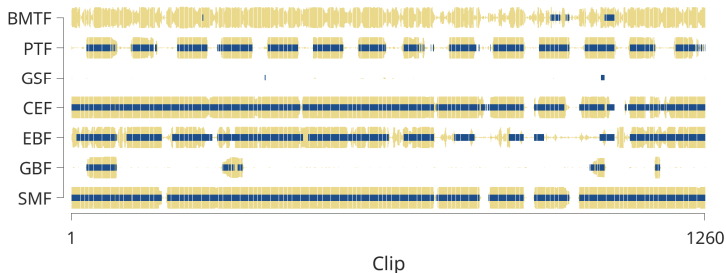
Environment Dynamics Model and the Truth

- The environmental model has a switching probability γ_f for frog f

$$\gamma_f \sim \text{uniform}(0, 1)$$

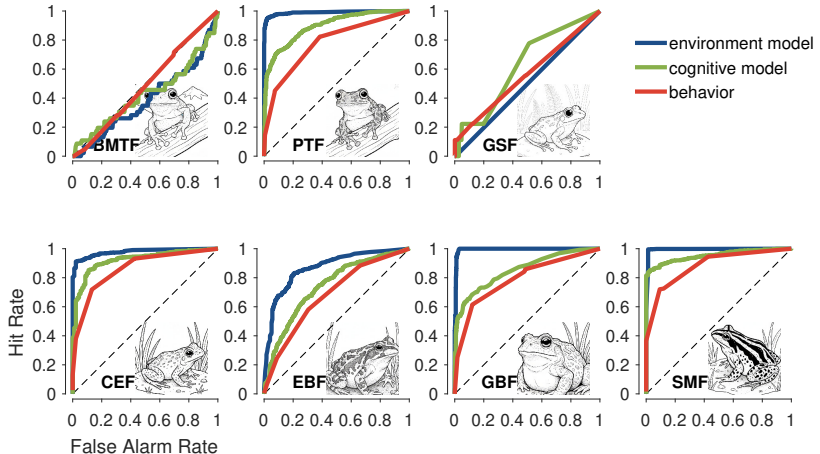
$$\tau_{f1} \sim \text{Bernoulli}\left(\frac{1}{2}\right)$$

$$\tau_{fk} \sim \begin{cases} \text{Bernoulli}(\gamma_f) & \text{if } \tau_{f(k-1)} = 1 \\ \text{Bernoulli}(1 - \gamma_f) & \text{if } \tau_{f(k-1)} = 0, k > 1 \end{cases}$$



Environment Dynamics Model Performance

- Performance is now near-perfect for three of the frog types, and better for the other two with behavioral signal



Discussion

Considering Environmental Structure

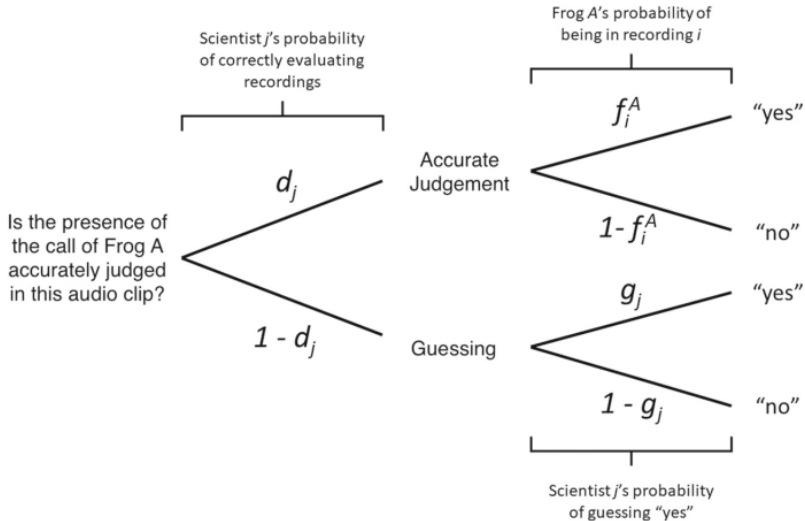
- It is easy as a cognitive modeler to focus too hard on internal cognitive processes
- The dynamics of the environment also influence observed behavior, and should be part of a complete model of the data-generating process
- Focusing on environmental regularities for this frog task could lead to improvement in
 - **model development**: more complicated environmental models are possible
 - e.g., model the covariances in the presence of frog types
 - **data collection**: presenting people with random clips corrupts the environmental structure their cognition may be able to use to make better judgments (Whitney, Sweeny, & Corbett, 2014)

Thanks!

References

- Brown, S. D., Sumnall, J., & Smith, P. R. (2024). Using a cognitive model to understand crowdsourced data from citizen scientists. *Behavior Research Methods*, 56(1), 421–434. doi: 10.3758/s13428-023-02289-w
- Lee, M. D. (2024). Using cognitive models to improve the wisdom of the crowd. *Current Directions in Psychological Science*, 33(5), 444–451. doi: 10.1177/09637214241264292
- Whitney, D., Sweeny, T. D., & Corbett, J. E. (2014). Visual ensemble statistics are represented as gradients of crowding. *Nature Neuroscience*, 17(1), 12–14. doi: 10.1038/nn.3603

Brown et al. (2024) Model



Brown et al. (2024) Results

